1 The complete pipeline

AetherGuard predicts, every morning, the chance of a rain-triggered landslide in every 5 km cell of the Himalaya (India, Nepal, Bhutan) and the Western Ghats (India) for today and the next three days. It first checks whether the rain sensors can be trusted, then reads the slope, and only then issues a level with a confidence flag and its reasons.

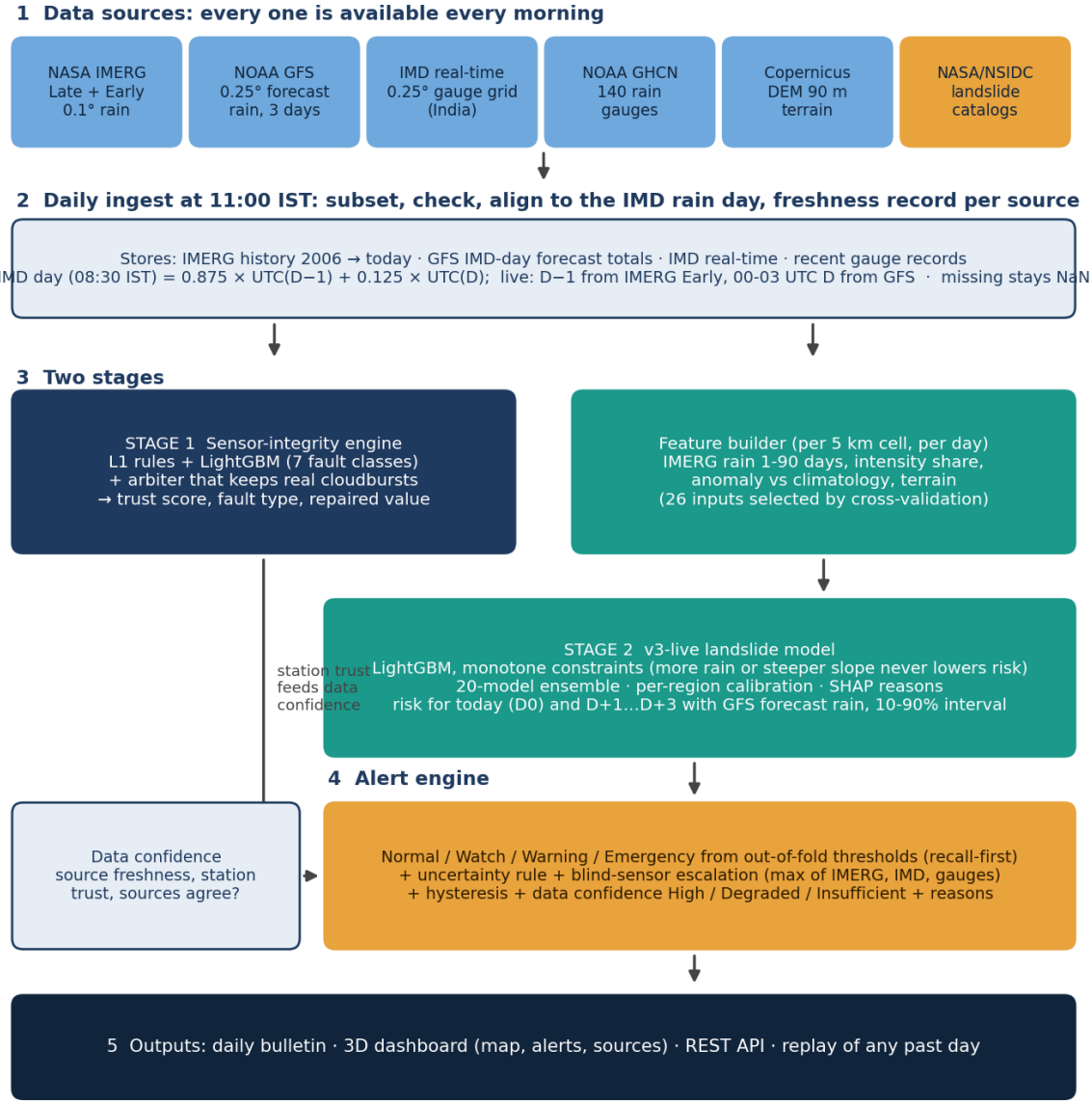


Figure 1. The live pipeline. Every input is published daily (IMERG about 14.5 h after the UTC day; GFS every 6 h; IMD real-time about 1 day; gauges a few days). The model is trained on the same products it receives live; for the newest day it receives the IMERG Early run, a gap measured in the operational backtest (page 3).

Step	When	What happens
Ingest	11:00 IST daily	Download the domain subset of each source, record its freshness, append to the history stores
Stage 1	same run	Score every recent gauge reading for faults; keep confirmed extremes; summarise station trust per cell
Features	same run	Build rain-history, rain-anomaly and terrain features per cell for D0; add GFS forecast rain for D+1…D+3
Stage 2	same run	Ensemble risk with a 10-90% interval; calibrated per region; top-5 reasons per alert
Alerts	same run	Levels with hysteresis; escalate when sensors are blind but any source shows heavy rain
Publish	same run	Bulletin, 3D dashboard (risk map, alerts with reasons, sensors and sources), API; a failed source is named, never hidden

2 Data the model works on, and how features are built

Training and live use read the same products (v3-live). The one difference, the IMERG Early run for the newest day, was measured by replaying 2018 as if live: see page 3.

Dataset	Role	Coverage used	In training	In live run
NASA GPM IMERG V07 (Late; Early for the same morning)	Model rain input	0.1°, 2006-10 → today, domain 8-36 °N, 72-98 °E	Yes (2007-2017)	Yes, daily
NOAA GFS 0.25° forecast	Rain for D+1…D+3	Archive 2021-2024 (skill); live latest cycle	Skill testing	Yes
IMD gridded rainfall (final / real-time)	Second opinion, data confidence	0.25°, India; final 2007-2024	v2 reference	Yes (if published)
NOAA GHCN-Daily	Stage 1 sensor checks	140 gauges (138 India, 2 Nepal)	Stage 1 training	Recent 120 days
Copernicus GLO-90 DEM	Terrain	495 tiles → 166,728 cells	Yes	Static
NASA GLC, COOLR, NSIDC HMA	Landslide labels	68,738 records → 919 day-exact landslide cell-days	Labels	—

Labels and splits

Rain-triggered landslides only, location accuracy ≤ 5 km, region by country (India, Nepal, Bhutan in the Himalaya; India in the Ghats). Frozen, checksummed splits: train 2007-2015, validation 2016-2017 (pooled for year-blocked cross-validation), unseen test 2018, window test 2023, case studies Kerala 2018 and Wayanad 2024. Non-events: 10 matched negatives per positive (same months, ≥ 3 cells and ±3 days from any recorded event) plus a weighted random sample of all cell-days to measure honest false-alarm rates.

Feature engineering (per 5 km cell, per IMD rain day D)

Group	Features	Why
Rain history	IMERG accumulations over 1, 3, 5, 7, 14, 30, 60 and 90 days; largest 1-day rain in 7/30 days; wet days in 30; days since 10 mm; 1-day share of the 3-day total	Short bursts and long soaking both trigger slides
Rain anomaly	3-, 7- and 30-day totals divided by the cell's own 2007-2015 climatology for that month	100 mm means different things in Cherrapunji and Ladakh
Wetness index (tested)	Antecedent precipitation index with decay 0.85, 0.90, 0.95; tested by cross-validation, not selected (no gain over rain history)	Wet soil fails sooner
Neighbourhood rain (tested)	Max and mean of 1- and 3-day rain over the 3 × 3 IMERG pixels; tested, not selected	Satellite storms are often displaced by a pixel
Terrain	Mean/std elevation, relief, mean/90th-percentile/max slope, curvature, aspect (sin, cos), wetness-index proxy	Steep, concave slopes concentrate water
Day basis	UTC products → IMD day: 0.875 × UTC(D−1) + 0.125 × UTC(D) (validated: mean error 1.85 mm vs exact hourly totals)	Rain must land on the right day

Each group enters the model only if year-blocked cross-validation shows it lowers false alarms at 90% detection by more than its standard error. Latitude, longitude, region and station presence are never inputs (they would learn where catalogs report landslides, not why slopes fail).

3 What the model can do, and how

Capability	How it does it
Daily risk for every 5 km cell, today and 3 days ahead	Each morning the newest IMERG rain is added to the 90-day history; D+1…D+3 add GFS forecast rain; the ensemble gives a probability and a 10-90% interval
Catch most landslide days while raising fewer false alarms	LightGBM learns how rain history, rain anomaly and terrain combine; monotone constraints keep it physically sensible; the Watch threshold is set from out-of-fold data so that ≥ 90% of past events would have been caught even after sampling noise
Know when its own rain gauges are lying	Stage 1 compares each gauge with its own history, neighbours and two independent rain grids; blocked, stuck, drifting and duplicated gauges are flagged; confirmed cloudbursts are kept
Never stay silent when blind	If a source is late, missing or disagrees while any source shows heavy rain, the level is raised and the bulletin says why
Explain every alert	Top-5 contributing factors (SHAP), the spread between rain sources, nearby gauge trust and the data-confidence flag
Replay any past day	The same code runs for a past date using only data available that morning (operational backtest)

Measured results

Result	Value	Source
Model, unseen year 2018: landslide days caught at Watch (v2 protocol model)	85.0% (77.5%-92.5%)	phase3_hazard_report
… share of cell-days on alert	21.4%	phase3_hazard_report
Best conventional rule at a similar catch (rainfall threshold)	32.4% on alert for 87.5% caught	phase3_hazard_report
Complete alert system, 2018: landslide days caught at Watch	91.2% (272 alerts per 1,000 cell-days)	phase4_alerts_report
Kerala floods, August 2018	23 of 23 landslide days flagged	phase4_alerts_report
Sensor faults caught (average over six fault types, unseen gauges)	0.70 vs 0.30 best rule check	phase2_integrity_report
v3-live (IMERG) cross-validated false alarms at 90% detection	19.6% vs 18.8% for v2	phase6_summary
v3-live verdict against v2	live-capable, not shown better than v2: paired difference in false alarms at 90% detection +0.7 points (95% CI -2.0 to +3.8), using satellite rain only	phase6_summary
v3-live, unseen year 2018 (single look): landslide days caught at Watch	83.8% (75.0%-91.2%) at 21.2% on alert; v2 85.0% at 21.4%	phase6_summary
Same model on live-timed inputs (2018 replayed as if live each morning)	82.1% of 56 landslide days caught vs 85.7% with archive inputs; scores rank-correlate 0.988	phase6_summary
Daily live operation on the laptop (11:00 IST)	7 live days published since 2026-10-16 (plus 7 backfilled); 12 of all 14 days with D0…D+3; failed sources named on 3	phase6_summary

Detection is measured on landslide days the model never saw; false alarms on a weighted random sample of all cell-days. Because the landslide catalogs are incomplete, false-alarm shares are upper bounds.

4 Why AetherGuard is better than existing systems

Existing landslide warnings in the region rely on fixed rainfall thresholds or susceptibility × rain rules fed by a single rain product. They fail in predictable ways. AetherGuard was designed around those failures and measured against the standard methods on the same data.

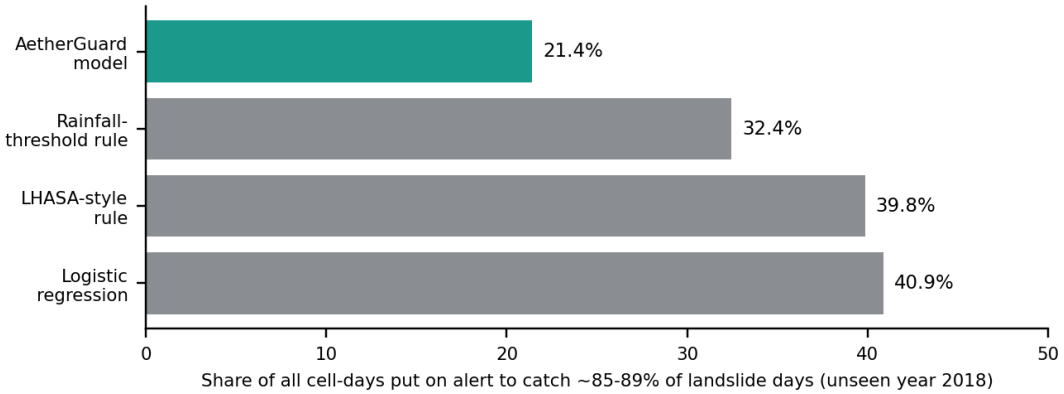


Figure 2. False alarms on the unseen year 2018, each method at its own pre-set Watch threshold (v2 protocol model; recall between 85% and 89% for all four).

Existing-system failure	Evidence	What AetherGuard does	Why it is not prone to it
Rain products miss local extremes. Global grids smooth an orographic cloudburst over 25 km.	Wayanad, 30 Jul 2024: IMD 179 mm, nearest trusted gauge 155 mm, ERA5 67 mm, CHIRPS 0 mm for the same day.	Uses several independent sources (IMERG, IMD real-time, gauges); the alert engine takes the maximum of the sources.	One wrong source cannot silence the warning: escalation uses the maximum, and disagreement lowers confidence instead of being averaged away.
Sensors fail silently. A clogged gauge reads 0 mm in a storm.	NOAA-style quality control caught 0.25 of injected faults on average; blocked-gauge faults 3-19%.	Stage 1 checks every reading against its own history, neighbours and two grids; arbiter keeps confirmed extremes.	Average detection 0.70; a flagged reading is never used to lower the rain estimate.
Fixed rain thresholds cry wolf.	Rainfall-threshold rule: 32.4% of cell-days on alert to catch 87.5%.	Learns from rain history, rain anomaly and terrain together; thresholds set by cross-validation with a recall-first margin.	21.4% of cell-days on alert for a similar catch on an unseen year (Figure 2).
Models trained on data that is not available live. Reanalysis or final gauge grids arrive weeks or a year late.	IMD final grid is yearly; CHIRPS final about 3 weeks; MERRA-2 weeks.	v3-live is trained on IMERG and terrain only, the same products the daily run receives.	Train/serve gap measured, not assumed: 2018 replayed with only the inputs that existed each morning caught 82.1% vs 85.7% with archive inputs.

5 Failures addressed, and why the pipeline is not prone to them

Existing-system failure	Evidence	What AetherGuard does	Why it is not prone to it
Rain lands on the wrong day. Products use UTC days; Indian gauges use 08:30 IST.	Gauge vs CHIRPS correlation peaks a day early (0.38 at lag −1 vs 0.16 at lag 0).	Every product is converted to the IMD rain day before any feature is built.	Tested: after alignment the match peaks on the same day; error 1.85 mm on average.
Inflated accuracy claims. Random splits and repeated test use.	Landslide catalogs are uneven (8 labelled days in 2007 vs 127 in 2015).	Frozen checksummed splits, year-blocked cross-validation for every choice, single frozen test look, bootstrap intervals.	Every reported number comes from a result file; protocol changes and reruns are disclosed.
Silent failure when blind. Systems show Normal when inputs are missing.	Operational feeds go down; IMD real-time was unreachable in one of three probes.	Data-confidence flag (High / Degraded / Insufficient) per cell; blind-sensor escalation; freshness of every source in the bulletin.	A missing source is named and lowers confidence; the run never publishes a silent Normal.

Built-in protections

- **Physics-consistent model:** more rain or a steeper slope can never lower the predicted risk (monotone constraints, checked with zero violations).
- **Recall-first operating point:** a missed landslide costs far more than a false alarm, so the Watch level is the highest threshold that still catches ≥ 90% of past events at the 5th bootstrap percentile.
- **Uncertainty is shown, not hidden:** a 20-model ensemble gives a 10-90% interval; its upper bound can raise Watch on its own.
- **Region-aware calibration:** the Himalaya and the Western Ghats get their own probability calibration, fixing the 3× under-prediction measured in the Ghats.
- **Every change is earned:** a feature group, model type or region rule is kept only if cross-validation shows a gain larger than its standard error.
- **Live freshness:** each source's issue time and latency are stored with every run, so a stale input is visible.
- **No rain, no alert:** a cell can reach Watch or above only after at least 11.5 mm of rain in 7 days (the 5th percentile of past landslide days); a level held without data steps down after 2 days.
- **Every alert audited:** each daily run checks every published cell for contradictions (risk inside its range, level matches its reasons, rain rule, confidence); 0 violations on 15 live days.

Known limits (stated, not hidden)

- Landslide catalogs are incomplete and rain-triggered only; the unseen test year has 80 labelled landslide days, so intervals are wide.
- Satellite rain at 0.1° still under-measures the most extreme orographic bursts; gauges and IMD real-time remain second opinions.
- Forecast days depend on rain that has not fallen. D+2 and D+3 now use the ECMWF ensemble (chosen over GFS on 2024 by a pre-declared rule; D+1 stays GFS). Each forecast cell is marked Firm when its level is the same with no further rain, the forecast and a wet case: 52-58% of cells, and these matched the level from observed rain 94-96% of the time (cells that depend on the forecast: 57-66%).
- In the monsoon the Watch level is broad: replaying 2018 as if live, 55.5% of monsoon cell-days were at Watch or above (Warning+ 22.4%, Emergency 8.7%); Watch caught 48 of 56 landslide days (51 before the rain trigger), Warning 40 and Emergency 30. Watch means heightened awareness; Warning and Emergency are the actionable levels.
- Research prototype, not validated for issuing public warnings.